IGCSE 0607 Practice Questions — E2.7

Sequences

Total Questions: 24 — **Total Marks:** 140 Marks

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1 Here are the first five terms of a number sequence.

7 11 15 19 23

(a) Write down the next term in this sequence.

..... **[1]**

(b) Find an expression, in terms of n, for the nth term of this sequence.

..... **[2]**

2 Here are the first four terms of a sequence.

5 11 17 23

(a) Find the next term of this sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

3 Here are the first four terms of a sequence.

5 12 19 26

(a) Write down the next term in the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

4 Here are the first four terms of a sequence.

5 11 17 23

(a) Find the next term in this sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

5 **(a)** Find the nth term of the sequence.

5 9 13 17 21

..... **[2]**

(b) Find the nth term of the sequence.

2 6 18 54 162

..... **[2]**

6 The first four terms of a sequence are

7 11 15 19

(a) Write down the next two terms of the sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

nth term = ... **[2]**

(c) Explain why 102 is not a term in this sequence.

..... **[2]**

7 **(a)** Sequence A is a linear sequence.

5 9 13 17 ...

(i) Find the next two terms of sequence A.

..... **[1]**

(ii) Find the nth term of sequence A.

..... **[2]**

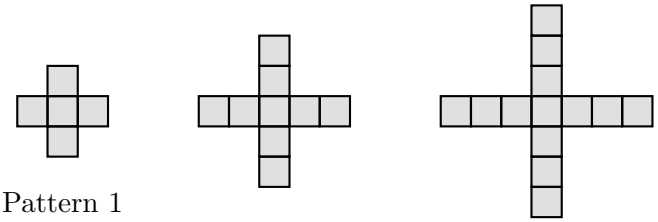
(b) Sequence B is a geometric sequence.

2 6 18 54 ...

Find the nth term of sequence B.

..... **[2]**

8 The diagram shows a sequence of patterns made from small squares.



Pattern 1

Pattern 2

Pattern 3

(a) Find the number of squares in Pattern 4.

..... **[1]**

(b) Find an expression, in terms of n, for the number of squares in Pattern n.

..... **[2]**

(c) Which pattern has exactly 141 squares?

Pattern ..... **[2]**

9 The first four terms of a sequence are

$$-1 \qquad 4 \qquad 13 \qquad 26$$

(a) Find an expression for the nth term of the sequence.

..... **[3]**

(b) Find the value of n when the nth term is 433.

$n =$ ..... **[3]**

10 The first five terms of a sequence, T_n, are given below.

4 14 34 68 120

(a) By finding the first, second and third differences, show that the nth term of the sequence is a cubic expression in n.

[2]

(b) Find an expression for T_n, the nth term of the sequence.

$T_n =$ ..... **[5]**

11 Here are the first five terms of a sequence.

3 9 17 27 39

Find an expression, in terms of n, for the nth term of this sequence.

..... **[3]**

12 Here are the first five terms of a sequence.

4 11 22 37 56

(a) Work out the next term of the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[3]**

13 The nth term of a sequence is $u_n = an^2 + bn$.

(a) Given that $u_1 = 5$ and $u_2 = 14$, write down two simultaneous equations in a and b.

.....

..... **[2]**

(b) Solve these simultaneous equations to find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

14 **(a)** The nth term of a sequence is $n^2 - 3n + 5$.
Work out the first three terms of this sequence.

1st term = ... , 2nd term = ... , 3rd term = ...
[2]

(b) The first four terms of another sequence are

2 9 20 35

Find an expression for the nth term of this sequence.

..... **[3]**

15 The first four terms of a sequence, T, are

2 9 22 41

(a) Find an expression for the nth term, T_n.

$T_n =$ ..... **[3]**

(b) A second sequence is given by the formula $U_n = 2^n + 3$.
Work out the value of U_5.

$U_5 =$ ..... **[1]**

(c) Find the value of n for which $U_n = 131$.

$n =$ ..... **[2]**

16 The first four terms of a sequence are

$$-1 \qquad 2 \qquad 9 \qquad 20$$

(a) Find the nth term of this sequence.

..... **[3]**

(b) The nth term of another sequence is given by $3^n - 2$.
Find the value of the term in this sequence that is closest to 1000.

..... **[3]**

17 The first five terms of a sequence are

2 9 20 35 54 ...

(a) Find the next term.

..... **[1]**

(b) Find the nth term of the sequence.

..... **[3]**

(c) The kth term of the sequence is 464.
Find the value of k.

$k =$ ..... **[3]**

18 Consider the following three sequences, P, Q and R.
Sequence P:

$$2 \quad 9 \quad 28 \quad 65 \quad 126 \quad \ldots$$

Sequence Q:

$$3 \quad 8 \quad 15 \quad 24 \quad 35 \quad \ldots$$

Sequence R:

$$\frac{2}{3} \quad \frac{9}{8} \quad \frac{28}{15} \quad \frac{65}{24} \quad \frac{126}{35} \quad \ldots$$

(a) Find the nth term of sequence P.

..... **[2]**

(b) Find the nth term of sequence Q.

..... **[2]**

(c) Write down the nth term of sequence R.

..... **[1]**

(d) Find the 12th term of sequence R, giving your answer as a fraction in its simplest form.

..... **[2]**

19 Here are the first five terms of a sequence.

2 9 28 65 126

(a) Find the next term of this sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

..... **[3]**

(c) The kth term of the sequence is 8001.
Find the value of k.

$k =$ ..... **[3]**

20 The table shows the first five terms of sequences A, B and C.

n	1	2	3	4	5
Sequence A	2	5	10	17	26
Sequence B	1	8	27	64	125
Sequence C	-1	3	17	47	99

(a) Find the nth term of sequence A.

..... **[2]**

(b) Find the nth term of sequence B.

..... **[1]**

(c) Use your answers to parts (a) and (b) to find an expression, in terms of n, for the nth term of sequence C.
Give your answer in its simplest form.

..... **[2]**

(d) Determine which term in sequence A has a value of 10001.

$n =$ ..... **[2]**

21 **(a)** The first five terms of a sequence are

$$-1 \quad 8 \quad 33 \quad 80 \quad 155$$

Use the method of differences to find an expression for the nth term of this sequence.

..... **[4]**

(b) An exponential sequence has nth term $P_n = 5 \times 3^{n-1}$.
The kth term of this sequence is 3645.
Work out the value of k.

$k =$ ..... **[2]**

(c) The nth term of sequence T is $T_n = \dfrac{2n-1}{3n+2}$.
Write down the value that T_n approaches as n becomes very large.

..... **[1]**

22 **(a)** A sequence has the following first four terms:

1 15 53 127

Use the method of differences to find an expression for the nth term.

nth term = ... **[4]**

(b) Another sequence, V, has its nth term defined by

$$V_n = \frac{n^2 + 1}{2^n}$$

(i) Find the values of V_1, V_2 and V_3.
Give each fraction in its simplest form.

V_1 = ... , V_2 = ... , V_3 = ... **[2]**

(ii) Work out the exact value of $V_5 - V_4$.

..... **[3]**

23 **(a)** The first five terms of a sequence are

4 5 14 37 80

Use the method of differences to find an expression for the nth term.

..... **[4]**

(b) The nth term of a different sequence is given by $T_n = a \times 2^n + b$.
The first term of this sequence is 10 and the third term is 34.

Find the values of a and b.

$a =$ ..... , $b =$ ..... **[4]**

(c) Calculate the 12th term of the sequence described in part (b).

..... **[2]**

24 The first five terms of three sequences, A, B and C, are shown below.

Sequence A:	5	9	13	17	21
Sequence B:	2	6	20	50	102
Sequence C:	$\frac{5}{2}$	$\frac{3}{2}$	$\frac{13}{20}$	$\frac{17}{50}$	$\frac{7}{34}$

(a) Find the nth term of sequence A.

..... **[2]**

(b) **(i)** Work out the 6th term of sequence B.

..... **[1]**

(ii) Find an expression for the nth term of sequence B.

..... **[4]**

(c) **(i)** Work out the 6th term of sequence C, giving your answer as a fraction in its simplest form.

..... **[1]**

(ii) Find an expression for the nth term of sequence C.

..... **[2]**